Test Series Code

13

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 15/04/2026

ANSWER KEY

1.	(2)	46.	(3)	91.	(4)	136.	(1)
2.	(3)	47.	(2)	92.	(3)	137.	(3)
3.	(1)	48.	(3)	93.	(1)	138.	(4)
4.	(1)	49.	(3)	94.	(4)	139.	(1)
5.	(1)	50.	(1)	95.	(2)	140.	(2)
6.	(4)	51.	(4)	96.	(3)	141.	(2)
7.	(4)	52.	(1)	97.	(2)	142.	(3)
8.	(1)	53.	(4)	98.	(1)	143.	(4)
9.	(2)	54.	(3)	99.	(3)	144.	(2)
10.	(2)	55.	(2)	100.	(4)	145.	(1)
11.	(4)	56.	(1)	101.	(3)	146.	(1)
12.	(4)	57.	(3)	102.	(1)	147.	(2)
13.	(4)	58.	(1)	103.	(2)	148.	(4)
14.	(3)	59.	(4)	104.	(3)	149.	(4)
15.	(1)	60.	(2)	105.	(4)	150.	(4)
16.	(2)	61.	(2)	106.	(2)	151.	(2)
17.	(3)	62.	(3)	107.	(1)	152.	(2)
18.	(1)	63.	(1)	108.	(1)	153.	(2)
19.	(2)	64.	(2)	109.	(4)	154.	(3)
20.	(3)	65.	(4)	110.	(3)	155.	(3)
21.	(2)	66.	(3)	111.	(2)	156.	(3)
22.	(2)	67.	(2)	112.	(4)	157.	(2)
23.	(1)	68.	(2)	113.	(4)	158.	(3)
24.	(2)	69.	(3)	114.	(3)	159.	(4)
25.	(3)	70.	(2)	115.	(3)	160.	(2)
26.	(3)	71.	(3)	116.	(3)	161.	(2)
27.	(3)	72.	(2)	117.	(2)	162.	(3)
28.	(3)	73.	(1)	118.	(3)	163.	(4)
29.	(3)	74.	(2)	119.	(1)	164.	(4)
30.	(4)	75.	(4)	120.	(4)	165.	(2)
31.	(2)	76.	(2)	121.	(2)	166.	(2)
32.	(3)	77.	(3)	122.	(3)	167.	(3)
33.	(1)	78.	(1)	123.	(1)	168.	(4)
34.	(3)	79.	(2)	124.	(3)	169.	(4)
35.	(3)	80.	(1)	125.	(2)	170.	(3)
36.	(4)	81.	(3)	126.	(3)	171.	(3)
37.	(3)	82.	(3)	127.	(1)	172.	(3)
38.	(4)	83.	(1)	128.	(2)	173.	(3)
39.	(2)	84.	(1)	129.	(3)	174.	(3)
40.	(3)	85.	(4)	130.	(1)	175.	(2)
41.	(2)	86.	(1)	131.	(4)	176.	(2)
42.	(2)	87.	(1)	132.	(1)	177.	(3)
43.	(4)	88.	(1)	133.	(4)	178.	(2)
44.	(3)	89.	(1)	134.	(3)	179.	(1)
45.	(2)	90.	(3)	135.	(2)	180.	(3)

[Contd...

Hints & Solutions

1. (2)

Radius of curvature $R \approx \frac{r^2}{2t} = \frac{(4\text{ cm})^2}{2(0.2\text{ cm})} = 40\text{ cm}$

Refractive index $\mu = \frac{c}{v} = \frac{3\times10^8}{2.4\times10^8} = 1.25$.

Using the lens maker's formula:

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R}\right) = (1.25-1)\left(\frac{1}{40}\right) = \frac{0.25}{40} = \frac{1}{160}$$

$\Rightarrow f = 160\text{ cm}$

(NCERT Ed. 2025-26, 12th, Page No. 234)

2. (3)

Equating the centripetal forces ($F_c = \frac{mv^2}{r}$):

$$\frac{m_{light}v_{light}^2}{r_{light}} = \frac{m_{heavy}v_{heavy}^2}{r_{heavy}}$$

$$\Rightarrow \frac{mv_L^2}{R} = \frac{4mv_H^2}{(R/4)}$$

Simplifying this gives: $\frac{v_L^2}{R} = \frac{16v_H^2}{R} \Rightarrow v_L^2 = 16v_H^2$

$\Rightarrow v_L = 4v_H$

Therefore, $k = 4$

(NCERT Ed. 2025-26, 11th, Page No. 63)

3. (1)

By conservation of angular momentum:

$$I_i\omega_i = I_f\omega_f \quad (3mR^2)\omega_0 = (3mR^2 + 2mR^2)\omega_f$$

$3mR^2\omega_0 = 5mR^2\omega_f \Rightarrow \omega_f = \frac{3}{5}\omega_0 = 0.6\omega_0$.

(NCERT Ed. 2025-26, 11th, Page No. 122)

4. (1)

Given $|a| = |v|$ at $x = 3\text{ cm}$ and $A = 5\text{ cm}$

Using standard SHM formulas: $\omega^2 x = \omega\sqrt{A^2 - x^2}$

Substituting the values:

$$\omega^2(3) = \omega\sqrt{5^2 - 3^2} \Rightarrow 3\omega = 4 \Rightarrow \omega = \frac{4}{3}\text{ rad/s}$$

The time period is $T = \frac{2\pi}{\omega} = \frac{2\pi}{4/3} = \frac{3\pi}{2}$ s

(NCERT Ed. 2025-26, 11th, Page No. 267)

5. (1)

Net force on the system is constant, so momentum is conserved.

$p_i = p_f$

$0 = m_1u_1 + m_2u_2$

$\frac{u_1}{u_2} = -\frac{m_2}{m_1} = -\left\lfloor\frac{1}{2}\right\rfloor$ as $\frac{m_1}{m_2} = \frac{2}{1}$

move in opposite direction with speed ratio 1:2.

(NCERT Ed. 2025-26, 11th, Page No. 101)

6. (4)

For Axis 1: $I_1 = m(0)^2 + m(l)^2 + m\left(\frac{l}{2}\right)^2$

$= ml^2 + \frac{1}{4}ml^2 = \frac{5}{4}ml^2$. (Correct)

For Axis 2:

$I_2 = m(0)^2 + m(0)^2 + m(\frac{\sqrt{3}}{2}l)^2 = \frac{3}{4}ml^2$. (Correct)

For Perpendicular Axis:

$I_\perp = m(0)^2 + m(l)^2 + m(l)^2 = 2ml^2$. (Correct)

(NCERT Ed. 2025-26, 11th, Page No. 114)

7. (4)

For a bright fringe, $d\sin\theta \approx d\theta = n\lambda$

$n\lambda = (0.1\times10^{-3}\text{ m})(0.018\text{ rad})$

$= 1.8\times10^{-6}\text{ m} = 1800\text{ nm}$

Testing integers for n to keep λ in the $[380, 740]$ range: If $n = 3, \lambda = 600\text{ nm}$

If $n = 4, \lambda = 450\text{ nm}$. Both fall within the visible range.

(NCERT Ed. 2025-26, 12th, Page No. 266)

8. (1)

Let the system accelerate with 'a' to the left. To keep m_2 from sliding, the tension T in the string must balance its weight: $T = m_2g$.

For m_1, the same tension T provides the necessary force to accelerate it at 'a' relative to the ground (since it doesn't move relative to M): $T = m_1a$.

Equating both: $m_1a = m_2g \Rightarrow a = \frac{m_2g}{m_1}$.

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9. (2)

By the Equation of Continuity,

$A_1v_1 = A_2v_2 \Rightarrow v \propto \frac{1}{A} \propto \frac{1}{d^2}$.

Minimum velocity occurs at the maximum diameter, and maximum velocity at the minimum diameter.

$\frac{v_{min}}{v_{max}} = \left(\frac{d_{min}}{d_{max}}\right)^2 = \left(\frac{3.0}{7.5}\right)^2 = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$.

(NCERT Ed. 2025-26, 11, Page No. 187)

10. (2)

Force $F = \text{Weight} + 2 \times T \times \text{circumference}$

(2 surfaces for a ring).

$T = 75 \text{ dyne/cm} = 0.075 \text{ N/m}$.

$F = 0.1 + (2 \times 0.075 \times 0.20) = 0.1 + 0.03 = 0.13 \text{ N}$

Given $\frac{1.17}{x} = 0.13 \Rightarrow x = \frac{1.17}{0.13} = 9$

(NCERT Ed. 2025-26, 11th, Page No. 194)

11. (4)

Since speed is constant, the upward driving force must balance all downward forces (gravity + friction).

Total Force

$F = Mg + f = (1200 \times 10) + 3000 = 15000 \text{ N}$.

Power

$P = F \times v = 15000 \times 2.5 = 37500 \text{ W} = 37.5 \text{ kW}$.

(NCERT Ed. 2025-26, 11th, Page No. 83)

12. (4)

The water equivalent (W) is the product of the mass (m) and the specific heat capacity (s) of the substance.

$W = m \times s = 40 \text{ g} \times 0.15 \text{ cal g}^{-1\circ}\text{C}^{-1} = 6 \text{ g}$.

(NCERT Ed. 2025-26, 11th, Page No. 208)

13. (4)

Initial Resistance $R = \rho\frac{l}{A}$.

New Resistance $R' = \rho\frac{2l}{A/2} = 4R$.

Initial Current $I = \frac{V}{R} = \frac{VA}{\rho l}$.

New Current $I' = \frac{V}{4R} = \frac{1}{4}\frac{VA}{\rho l}$.

(NCERT Ed. 2025-26, 12th, Page No. 84)

14. (3)

Secondary Voltage (V_s):

$V_s = V_p\left(\frac{N_s}{N_p}\right) = 200 \times \left(\frac{50}{1000}\right) = 10 \text{ V}$. $\rightarrow$ (A)-(II)

Secondary Current (I_s): $I_s = \frac{V_s}{R_s} = \frac{10}{2} = 5 \text{ A}$. $\rightarrow$ (B)-(I)

Primary Current (I_p): For an ideal transformer,

$I_p = I_s\left(\frac{N_s}{N_p}\right) = 5 \times \left(\frac{50}{1000}\right) = 0.25 \text{ A}$. $\rightarrow$ (C)-(IV)

Output Power (P_{out}): $P_{out} = V_s \times I_s = 10 \times 5 = 50 \text{ W}$.

$\rightarrow$ (D)-(III)

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15. (1)

$AB\cos\theta = AB\sin\theta \Rightarrow \tan\theta = 1 \Rightarrow \theta = 45^\circ$

$\left|\vec{A} - \vec{B}\right| = \sqrt{A^2 + B^2 - 2AB\cos 45^\circ}$

$= \sqrt{A^2 + B^2 - \sqrt{2}AB}$

(NCERT Ed. 2025-26, 11th, Page No. 102)

16. (2)

From energy conservation: $E_{3\to1} = E_{3\to2} + E_{2\to1}$.

Using $E = \frac{hc}{\lambda}$, we get: $\frac{1}{\lambda_1} = \frac{1}{\lambda_3} + \frac{1}{\lambda_2} = \frac{\lambda_2 + \lambda_3}{\lambda_2\lambda_3}$.

Rearranging: $\lambda_1 = \frac{\lambda_2\lambda_3}{\lambda_2 + \lambda_3}$

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17. (3)

Heat current

$H_1 = H_2 \Rightarrow \frac{K_1A_1(450 - T)}{L_1} = \frac{K_2A_2(T - 0)}{L_2}$

$\frac{450 - T}{T} = \frac{K_2}{K_1} \times \frac{A_2}{A_1} \times \frac{L_1}{L_2} = 9 \times \frac{1}{2} \times 2 = 9$

$450 - T = 9T$

$\Rightarrow T = 45^\circ\text{C}$

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18. (1)

A negative charge moving towards the observer is equivalent to a conventional current moving away from the observer.

By the right-hand thumb rule, if the thumb points away from you, the fingers curl in the clockwise direction.

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19. (2)

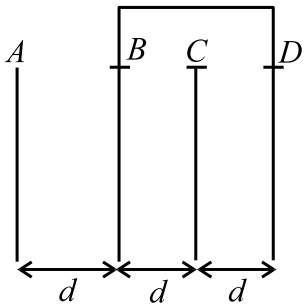


Plate A is terminal 1; Plate C is terminal 2; B-D are connected.

The system is C_{AB} in series with (C_{BC} parallel C_{CD}).

$$C_{eq} = \text{series of } C \text{ and } 2C = \frac{2}{3}C$$

$$C = \frac{\epsilon_0 Area}{d} = \frac{\epsilon_0 (2\times 1.5)}{d} = \frac{3\epsilon_0}{d}$$

$$C_{eq} = \frac{2}{3}\cdot\frac{3\epsilon_0}{d} = \frac{2\epsilon_0}{d} \Rightarrow x = 2$$

(NCERT Ed. 2025-26, 12th, Page No. 72)

20. (3)

Time to collide: $t = \dfrac{\text{Relative Distance}}{\text{Relative Velocity}} = \dfrac{150}{150} = 1\text{ s}$.

Collision height from ground

$$= 150 - \frac{1}{2}(10)(1)^2 = 145\text{ m}.$$

Velocities just before collision: Block $v_1 = -10$ m/s ; Bullet $v_2 = 150 - 10 = 140$ m/s .

Conservation of Momentum:

$$0.06(-10) + 0.04(140) = (0.06 + 0.04)V$$

$$\Rightarrow -0.6 + 5.6 = 0.1V \Rightarrow V = 50\text{ m/s (upwards)}.$$

Further height reached $h' = \dfrac{V^2}{2g} = \dfrac{50^2}{20} = 125\text{ m}$.

Total height from ground $= 145 + 125 = 270$ m.

Height above tower $= 270 - 150 = 120$ m.

(NCERT Ed. 2025-26, 11th, Page No. 84)

21. (2)

By the work-energy theorem, the work done by the tangential force equals the change in kinetic energy:

$$W = \Delta KE$$

$$(ma_t)\times(n\cdot 2\pi r) = KE$$

$$a_t = \frac{KE}{m\cdot n\cdot 2\pi r} = \frac{10\times 10^{-3}}{0.02\times 5\times 2\pi\times(0.1/\pi)}$$

$$= \frac{0.01}{0.02\times 1} = 0.5\text{ m/s}^2$$

(NCERT Ed. 2025-26, 11th, Page No. 76)

22. (2)

Slicing the magnet longitudinally divides its cross-sectional area evenly but keeps the length unchanged.

Therefore, the new pole strength is $m' = \dfrac{m}{3}$.

The new magnetic moment is

$$M' = m'\times(2L) = \left(\frac{m}{3}\right)\times(2L) = \frac{M}{3}.$$

(NCERT Ed. 2025-26, 12th, Page No. 137)

23. (1)

For heights much smaller than the radius ($h \ll R$), the acceleration due to gravity is $g_h \approx g\left(1 - \dfrac{2h}{R}\right)$.

The fractional decrease is $\dfrac{\Delta g}{g} = \dfrac{2h}{R}$

Given a 3% decrease: $0.03 = \dfrac{2h}{6400}$

$$\Rightarrow h = \frac{0.03\times 6400}{2} = 96\text{ km}$$

(NCERT Ed. 2025-26, 11th, Page No. 133)

24. (2)

Area $A = \pi r^2 \Rightarrow \dfrac{\Delta A}{A} = 2\dfrac{\Delta r}{r}$

The lateral/radial strain is $\dfrac{\Delta r}{r} = \dfrac{-2.4\%}{2} = -1.2\%$

Poisson's ratio is $\sigma = -\dfrac{\Delta r / r}{\Delta L / L}$

Longitudinal strain $\dfrac{\Delta L}{L} = -\dfrac{-1.2\%}{0.4} = 3.0\%$

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25. (3)

The electric field due to an infinite line charge is $E = \frac{2k\lambda}{r}$

$\lambda = \frac{E \cdot r}{2k} = \frac{3.6\times10^5 \times 0.05}{2\times9\times10^9} = \frac{0.18\times10^5}{18\times10^9}$

$= 10^{-6}$ C/m $= 1\ \mu$C/m

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26. (3)

The time gap between crossing a specific height h is given by $\Delta t = \sqrt{T^2 - \frac{8h}{g}}$, where T is the total time of flight.

Squaring both sides: $T^2 = (\Delta t)^2 + \frac{8h}{g}$

$T^2 = (4)^2 + \frac{8(10)}{10} = 16 + 8 = 24$

$\Rightarrow T = \sqrt{24}$ s

(NCERT Ed. 2025-26, 11th, Page No. 19)

27. (3)

A silvered slab acts like a plane mirror shifted forward by the apparent depth of the glass. The apparent distance of the final image from the front face is given by $v_{front} = u + \frac{2t}{\mu}$.

The apparent distance behind the physical silvered rear face is $v_{back} = v_{front} - t = u + \frac{2t}{\mu} - t$.

Given $v_{back} = 19$, $u = 15$, and $t = 12$:

$19 = 15 + \frac{24}{\mu} - 12 \Rightarrow 19 = 3 + \frac{24}{\mu}$

$\Rightarrow 16 = \frac{24}{\mu} \Rightarrow \mu = 1.5$

(NCERT Ed. 2025-26, 12th, Page No. 229)

28. (3)

To completely enclose a charge placed on an edge, we need to join four identical cubes together at that specific edge.

Gauss's Law: The total flux through the entire system of 4 cubes is $\Phi_{total} = \frac{q}{\varepsilon_0}$.

Since the flux is distributed equally among the 4 cubes, the flux through a single cube is:

$\Phi = \frac{1}{4} \times \frac{q}{\varepsilon_0} = \frac{q}{4\varepsilon_0}$

(NCERT Ed. 2025-26, 12th, Page No. 30)

29. (3)

Thick copper wire A has almost zero resistance compared to manganin wire B.

In parallel, current takes the path of least resistance.

Thus, all 2 A current flows through A, and zero flows through B.

The total 2 A then flows through series wire C.

Currents $(A, B, C) = 2$ A, 0, 2 A

(NCERT Ed. 2025-26, 12th, Page No. 84)

30. (4)

The given combination of gates corresponds to AND gate, so the output will be 1 when both the inputs are 1.

31. (2)

The graph is a straight line following Einstein's equation: $KE_{max} = h\nu - \phi$.

Since the graph is a straight line, KE_{max} depends linearly on frequency ν.

(NCERT Ed. 2025-26, 12th, Page No. 282)

32. (3)

$v_{max} = A\omega = A\sqrt{\frac{K}{m}}$

Given $v_{\underset{A}{max}} = v_{\underset{B}{max}} \Rightarrow A_1\sqrt{\frac{K_1}{m}} = A_2\sqrt{\frac{K_2}{m}}$

$\Rightarrow \frac{A_1}{A_2} = \sqrt{\frac{K_2}{K_1}}$

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[Contd...

33. (1)

At position A (steady state), current $I_0 = \frac{E}{R}$.

Energy stored in inductor $U_L = \frac{1}{2}LI_0^2$.

At position B, this energy is transferred to the capacitor. Max charge q occurs when all energy is in the capacitor: $\frac{1}{2}LI_0^2 = \frac{q^2}{2C}$.

$q = \sqrt{LC} \times I_0 = \left(\sqrt{LC}\right)\frac{E}{R}$

(NCERT Ed. 2025-26, 12th, Page No. 189)

34. (3)

Floating condition: $g = \omega^2 R \Rightarrow \omega = \sqrt{\frac{g}{R}}$.

Time period $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{R}{g}}$.

$T = 2 \times 3.14 \times \sqrt{\frac{6400 \times 10^3}{10}}$

$\approx 5024 \text{ s} \approx 84.6 \text{ min}$

(NCERT Ed. 2025-26, 11th, Page No. 133)

35. (3)

Heat in A (Isobaric): $Q = nC_P T_0$ (since the piston is free to move).

Heat in B (Isochoric): $Q = nC_V \Delta T_B$ (since the piston is fixed).

Equating Heat: $nC_P T_0 = nC_V \Delta T_B \Rightarrow \Delta T_B = \left(\frac{C_P}{C_V}\right)T_0$

For monatomic gas, $\frac{C_P}{C_V} = \gamma = \frac{5/2R}{3/2R} = \frac{5}{3}$

Thus, $\Delta T_B = \frac{5}{3}T_0$

Both statements are true and the reason explains the assertion.

(NCERT Ed. 2025-26, 11th, Page No. 253)

36. (4)

Magnetic field $\vec{B}$ is parallel to the x-axis. It does not change the speed in the yz-plane ($v_{yz} = v_0$).

Electric field $\vec{E}$ accelerates the particle along the x-axis: $v_x = \frac{qE_0}{m}t$

Net speed $v = \sqrt{v_x^2 + v_0^2}$

To double the speed ($v = 2v_0$):

$(2v_0)^2 = v_x^2 + v_0^2 \Rightarrow v_x = \sqrt{3}v_0$

Substitute v_x: $\frac{qE_0 t}{m} = \sqrt{3}v_0 \Rightarrow t = \frac{\sqrt{3}mv_0}{qE_0}$

(NCERT Ed. 2025-26, 12th, Page No. 112)

37. (3)

Substitute the standard dimensional formulas for each quantity:

$[E] = ML^2T^{-2}$, $[L] = ML^2T^{-1}$, $[m] = M$,

and $[G] = M^{-1}L^3T^{-2}$.

$\left[\frac{EL^2}{m^4G^2}\right] = \frac{\left(ML^2T^{-2}\right)\cdot\left(M^2L^4T^{-2}\right)}{M^4\cdot\left(M^{-2}L^6T^{-4}\right)}$

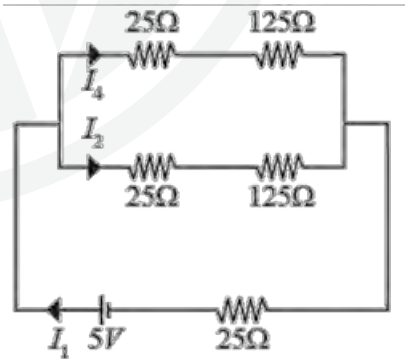
$= \frac{M^3L^6T^{-4}}{M^2L^6T^{-4}} = M^1$

This leaves only M, which is the dimension of Mass.

(NCERT Ed. 2025-26, 11th, Page No. 7)

38. (4)

As D_2 is reverse biased. So reduced circuit.



$R_{\text{Total}} = \frac{150 \times 150}{300} + 25 = 100\Omega$

$I_1 = \frac{5}{100} = 0.05A$

Using KCL $I_1 = I_2 + I_4$

$I_2 = I_4 = \frac{0.05}{2} = 0.025A$

$\frac{I_1}{I_2} = \frac{0.05}{0.025} = 2$

(NCERT Ed. 2025-26, 12th, Page No. 337)

[Contd...

39. (2)

Identify the positions of the individual centers of mass:

Let the center of the disc be at the origin $(0,0)$.

Mass $m_1 = 2M$.

Since the ring has radius R and passes through the center of the disc, its own center must be at a distance from the origin. Let its center be at $(R,0)$

Mass $m_2 = M$.

Apply the Center of Mass formula:

$$X_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$X_{cm} = \frac{(2M \times 0) + (M \times R)}{2M + M}$$

$$X_{cm} = \frac{MR}{3M} = \frac{R}{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 96)

40. (3)

From the standard wave equation, the wave number

$$k = 0.314 \approx \frac{\pi}{10}$$

Wavelength $\lambda = \frac{2\pi}{k} = \frac{2\pi}{\pi/10} = 20$ m

For the 5th harmonic,

Length $L = \frac{5\lambda}{2} = \frac{5 \times 20}{2} = 50$ m.

(NCERT Ed. 2025-26, 11th, Page No. 291)

41. (2)

According to the Work-Energy Theorem, the force F is equal to the rate of change of kinetic energy with position: $F = \frac{dK}{dx}$ (slope of K - x graph).

At $x = 9$ m, the particle is on the downward slope.

Slope $= \frac{\Delta K}{\Delta x} = \frac{0-20}{10-6} = \frac{-20}{4} = -5$ N.

The magnitude of the force is 5 N.

(NCERT Ed. 2025-26, 11th, Page No. 76)

42. (2)

Average energy density $U = \frac{B_0^2}{2\mu_0}$.

$$B_0 = \sqrt{2\mu_0 U} = \sqrt{2\left(4\pi \times 10^{-7}\right)\left(\frac{2}{\pi} \times 10^{-7}\right)}$$

$$B_0 = \sqrt{16 \times 10^{-14}} = 4 \times 10^{-7} \text{ T} = 400 \text{ nT}$$

(NCERT Ed. 2025-26, 12th, Page No. 214)

43. (4)

The entire kinetic energy of the gas converts into its internal energy.

For n moles, $K = \frac{1}{2}(nM)(3v)^2 = \frac{9}{2}nMv^2$.

The change in internal energy is

$$\Delta U = nC_v \Delta T = n\frac{R}{\gamma - 1}\Delta T$$.

Equating the two:

$$n\frac{R}{\gamma - 1}\Delta T = \frac{9}{2}nMv^2 \Rightarrow \Delta T = \frac{9(\gamma - 1)Mv^2}{2R}$$

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44. (3)

We know thas mass

$$m = \rho \times \frac{4}{3}\pi R^3$$

$\therefore R \propto m^{\frac{1}{3}}$ ($\because$ Density is equal)

Weight $= W \propto g \propto \frac{m}{R^2}$

$$\therefore W \propto \frac{m}{m^{\frac{2}{3}}} \propto m^{\frac{1}{3}}$$

$$\frac{W}{W_1} = \left(\frac{m}{2m}\right)^{\frac{1}{3}} = \left(\frac{1}{2}\right)^{\frac{1}{3}}$$

So, $W_1 = 2^{\frac{1}{3}}W$

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45. (2)

The original frequency of the string is $f = 440 \pm 6$, meaning it is either 446 Hz or 434 Hz.

Increasing the tension raises the string's frequency. If it were 446 Hz, an increase (e.g., to 448 Hz) would raise the beat frequency ($448 - 440 = 8$ Hz), which matches the observation. If it were 434 Hz, an increase would lower the beat frequency.

Thus, $f_{string} = 446$ Hz.

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[Contd...

46. (3)

First ionization enthalpy increases across a period and decreases down a group; nitrogen's stable half-filled $2p^3$ configuration makes it highest.

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47. (2)

Molarity = 3.2 M

Let volume of solution = 1000 ml = volume of solvent

Mass of solvent = 1000 × 0.4 = 400 gm

N_{solute} = 3.2 mole

$$\text{Molality (m)} = \frac{3.2}{400/1000} = 8$$

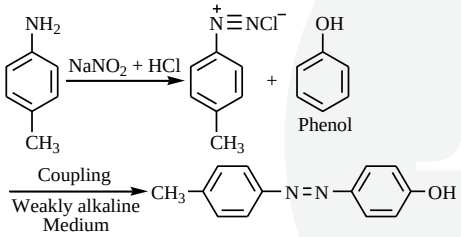
(NCERT Ed. 2025-26 12th Part-I Page No. 03)

48. (3)

Butyne does not have terminal alkyne group

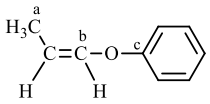
(NCERT Ed. 2025-26 11th Part-II Page No. 316)

49. (3)



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50. (1)



$a = sp^3, b = sp^2, c = sp^2$

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51. (4)

The **correct** order of basic nature in aqueous solution is 2° > 3° > 1°

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52. (1)

$-OCH_3$ and $-NHCOCH_3$ activate ring; –CN and $-SO_3H$ are meta directors; activating groups generally direct ortho/para; halides deactivate.

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53. (4)

Sulphanilic acid has $-SO_3H$, picric acid has phenolic –OH, only aspirin contains –COOH.

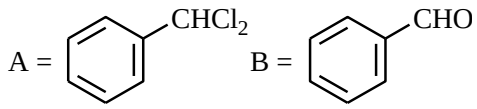
(NCERT Ed. 2025-26 12th Part-II Page No. 244)

54. (3)

Nonmetals are more electronegative, metal–nonmetal compounds are usually ionic, while nonmetal oxides are acidic and metal oxides are basic generally.

(NCERT Ed. 2025-26 11th Part-I Page No. 91)

55. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 233)

56. (1)

Covalent radius and ionic radius increase regularly down group-17, while electron affinity and ionization energy display expected fluorine irregularities.

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57. (3)

Primary amines give carbylamine test and diazotization; Hinsberg reagent is benzenesulfonyl chloride; tertiary amines do not react easily here.

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58. (1)

Pure arylamines are usually colourless. On storage, air causes oxidation, producing coloured impurities.

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59. (4)

Ribose is a D-pentose and free ribose is reducing. In nucleic acids it prefers furanose, not pyranose; α form incorrect.

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60. (2)

BrF_5 and $XeOF_4$ each have six electron domains around central atom, so electron-pair geometry is octahedral though molecular shape differs.

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61. (2)

Protein monomers are α-amino acids. Hence liberated amino acids have amino group on carbon adjacent to carboxyl group, the α-carbon.

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[Contd...

62. (3)

Te lies on the metalloid boundary and exhibits intermediate electrical and chemical behavior, unlike clearly metallic Sc, Pb, and Bi.

63. (1)

Mn is $[Ar]3d^5 4s^2$; p-electrons total 12 and d-electrons total 5.

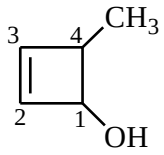
(NCERT Ed. 2025-26 11th Part-I Page No. 65)

64. (2)

Extensive properties depend on amount of substance; volume changes with system size, unlike the others.

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65. (4)



Alcohol has been given priority over alkene

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66. (3)

Water and ethanol have unlike intermolecular interactions significantly different from like interactions, causing non-ideal behaviour.

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67. (2)

Poor shielding by d-electrons makes Ga slightly smaller than Al; size otherwise increases down the group.

68. (2)

Spontaneity is guaranteed when ΔH is negative and ΔS is positive, making ΔG negative at all temperatures.

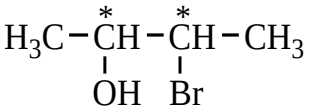
(NCERT Ed. 2025-26 11th Part-I Page No. 157)

69. (3)

Concentration falls from 0.16 to 0.02, an eightfold decrease, corresponding to three half-lives or 45 minutes.

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70. (2)



$$\begin{bmatrix} R\ R \\ R\ S \\ S\ S \\ S\ R \end{bmatrix}$$

Unsymmetrical n = 2

Total stereoisomer = $2^n = 2^2$

Total stereoisomer = 4

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71. (3)

Carbon dioxide is a non-metal oxide and forms carbonic acid in water, showing acidic nature.

72. (2)

Cadmium sulphide is characteristically yellow, whereas zinc sulphide is white and lead sulphide is black.

73. (1)

Option (1) has resonance stabilised benzylic carbocation.

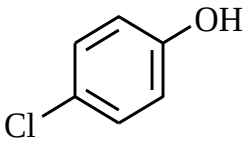
(NCERT Ed. 2025-26 11th Part-II Page No. 276)

74. (2)

Copper exception, negative m values, and Aufbau are correct; total nodes equal $n - 1$, not $n + \ell$.

(NCERT Ed. 2025-26 11th Part-I Page No. 57)

75. (4)



EWG increases acidity

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76. (2)

Gold shows anomalous configuration because filled 5d subshell gains extra stability, giving $5d^{10}6s^1$.

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[Contd...

77. (3)

Leclanche cell	Reaction at anode $Zn \rightarrow Zn^{2+} + 2e^-$
Ni-Cd cell	Rechargeable
Fuel cell	Converts energy of combustion into electrical energy
Mercury cell	Does not involve any ion in solution and is used in hearing aids

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78. (1)

Using $p = K_A X$, $= \frac{p}{KH} = \frac{0.05}{2.5\times10^4} = 2\times10^{-6}$.

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79. (2)

(I) Second order

(II) First order

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80. (1)

For zero potential, $P_{H_2} = [H^+]^2$; in pure water $[H^+] = 10^{-7}$, so pressure is 10^{-14} bar.

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81. (3)

P = n-factor of $Ca(OH)_2$ = 2

Q = n-factor of HCl = 1

So, P + Q = 2 + 1 = 3

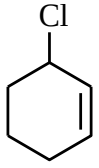
(NCERT Ed. 2025-26 11th Part-II Page No. 249)

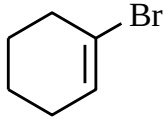
82. (3)

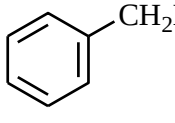
- Enzymes are very specific for their action.
- Enzymes are needed only in small quantities for the progress of a reaction.

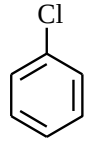
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83. (1)

 : Allylic halide

 : Vinylic halide

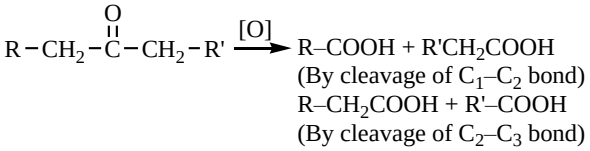
 : Benzylic halide

 : Aryl halide

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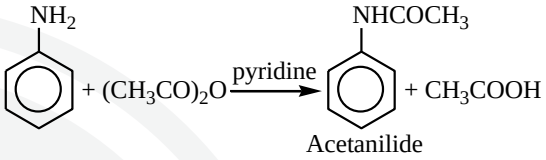
84. (1)

Possible products on oxidation are



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85. (4)



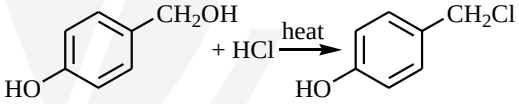
(NCERT Ed. 2025-26 12th Part-II Page No. 272)

86. (1)

In metal displacement reactions, a metal in a compound can be displacement by another metal in the uncombined state.

(NCERT Ed. 2025-26 12th Part-I Page No. 242)

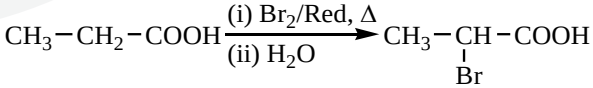
87. (1)



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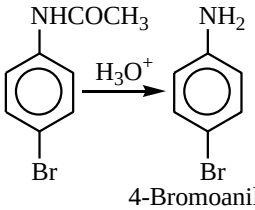
88. (1)

Hell-Volhard-Zelinsky reaction



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89. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 272)

90. (3)

Moles of N_2 = 2 mole

Moles of H_2 = 6 mole

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[Contd...

91. (4)

In a typical test cross an organism showing a dominant phenotype is crossed with the recessive parent instead of self-crossing. The progenies of such a cross can easily be analysed to predict the genotype of the test organism. Snapdragon (*Antirrhinum* sp.) is a classic example of incomplete dominance. In snapdragon, when a homozygous red-flowered plant (RR) is crossed with a homozygous white-flowered plant (rr), the F_1 progeny (Rr) are pink, not red. A test cross involves crossing the F_1 progeny (Rr) with the homozygous recessive parent (rr). The cross Rr (Pink) × rr (White) results in a 1:1 ratio of pink flowered and white flowered plants.

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92. (3)

Certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO_2 and H_2 (gobar gas). These bacteria are collectively called methanogens, and one such common bacterium is *Methanobacterium*. These bacteria are commonly found in the anaerobic sludge during sewage treatment. These bacteria are also present in the rumen (a part of stomach) of cattle. A lot of cellulosic material present in the food of cattle is also present in the rumen.

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93. (1)

Chara	Green alga	Chloroplasts contain pyrenoids, chlorophyll *a* and *b*
Anabaena	Cyanobacterium	Chloroplasts are not present, can fix atmospheric nitrogen in specialised cells
Polysiphonia	Red alga	Oogamous sexual reproduction, chloroplasts contain *a* and *d*
Dictyota	Brown alga	Stored food is laminarin or mannitol, chloroplasts contain *a* and *c*

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94. (4)

Sexual mode of reproduction enables creation of new variants, so that survival advantage is enhanced. Clones are produced by asexual reproduction.

(NCERT Ed. 2025-26, 12th, Page No. 1)

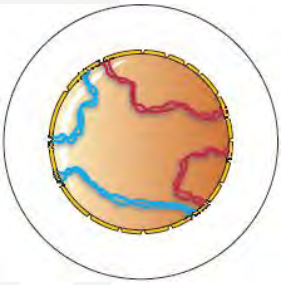
95. (2)

While both have pigments, *Euglena* lacks a cell wall (it has a protein-rich layer called a pellicle), whereas the Venus fly trap has a cellulosic cell wall. Both are eukaryotic, but *Euglena* is unicellular, while the Venus fly trap is multicellular. *Euglena* is aquatic and shows locomotion via flagella. The Venus fly trap is a terrestrial plant and does not show locomotion.

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96. (3)

The cell in the image represents Early Prophase of mitosis. The DNA has already replicated but the daughter chromosomes are yet to be formed (in anaphase). It is the first stage of the most dramatic phase of cell cycle (M phase). The cell shown here is an animal cell since it lacks cell wall, and therefore will undergo cytokinesis by the formation of cell furrow.



Early Prophase

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97. (2)

Careful analysis of records shows that extinctions across taxa are not random; some groups like amphibians appear to be more vulnerable to extinction. Presently, 12 per cent of all bird species, 23 per cent of all mammal species, 32 per cent of all amphibian species and 31 per cent of all gymnosperm species in the world face the threat of extinction. The largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth- it is home to more than 40,000 species of plants, 3,000 of fishes, 1,300 of birds, 427 of mammals, 427 of amphibians, 378 of reptiles and of more than 1,25,000 invertebrates. The number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals. Western Ghats have a greater amphibian species diversity than the Eastern Ghats.

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[Contd...

98. (1)

The interphase lasts more than 95% of the duration of cell cycle. The cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle. Cells in this stage remain metabolically active but no longer proliferate unless called on to do so depending on the requirement of the organism. In the S phase, the amount of DNA doubles but the chromosome number remains the same. The M Phase starts with karyokinesis and ends with cytokinesis.

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99. (3)

The pteridophytes are the first terrestrial plants to possess vascular tissues – xylem and phloem. Bryophytes lack true roots, stem or leaves.

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100. (4)

Trees occupy top vertical strata or layer of a forest, shrubs the second and herbs and grasses occupy the bottom layers.

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101. (3)

Since both leaf A and B increased their surface area by the same total amount 5 cm^2, in the same time period, both have same absolute growth rate i.e. 5 cm^2 per day. Their relative growth rate is 0.1 and 0.05, respectively.

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102. (1)

Conjoint open vascular bundles	Dicot stem
Radial open vascular bundles	Dicot root
Conjoint closed vascular bundles	Monocot stem
Radial closed vascular bundles	Monocot root

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103. (2)

Pedigree analysis is a method used to study the inheritance of Mendelian traits in human families over several generations. Phenylketonuria is an autosomal recessive Mendelian disorder. Polygenic inheritance refers to traits controlled by three or more genes (like human height or skin color) where the phenotype is the result of the cumulative effect of all the genes. Since ABO blood grouping is controlled by a single gene, it is not polygenic inheritance. It is a case of multiple alleles.

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104. (3)

Mesosomes help in cell wall formation, DNA replication and distribution to daughter cells. They also help in respiration, secretion processes, to increase the surface area of the plasma membrane and enzymatic content.

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105. (4)

The curly top virus spreads through a plant via the phloem (a part of conducting tissue system). It is a complex tissue made of different types of cells (most of them are living).

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106. (2)

Schleiden and Schwann together formulated the cell theory. Rudolf Virchow (1855) first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e cellula*). An improved model of the structure of cell membrane was proposed by Singer and Nicolson (1972) widely accepted as fluid mosaic model. Robert Brown discovered the nucleus.

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107. (1)

In non-cyclic photophosphorylation (the Z-scheme), electrons travel from Photosystem I to Ferredoxin and then to FNR to reduce $NADP^+$ into NADPH. In cyclic photophosphorylation, electrons are cycled back from Ferredoxin to the Cytochrome B6f complex; therefore, FNR is not involved in the cyclic pathway. The stroma lamellae membranes possess PS I but lack PS II and the NADP reductase enzyme (FNR). FNR is not the primary electron acceptor from PS II, but is associated with the end of the electron transport chain after Photosystem I.

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108. (1)

In most ecosystems, all the pyramids, of number, of energy and biomass are upright, i.e., producers are more in number and biomass than the herbivores, and herbivores are more in number and biomass than the carnivores.

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109. (4)

During transcription, adenosine complements forms base pair with uracil instead of thymine. However, unlike in the process of replication, which once set in, the total DNA of an organism gets duplicated, in transcription only a segment of DNA and only one of the strands is copied into RNA.

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110. (3)

The first stable product of CO_2 fixation in C_4 plants is oxaloacetic acid (OAA).

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111. (2)

Pea - Legume
Coconut - Drupe
Sunflower - Cypsela

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112. (4)

Formation of the female gametophyte: This is a pre-fertilization event. The megaspore undergoes mitoses to form the 7-celled, 8-nucleate embryo sac (female gametophyte) within the ovule. Transfer of pollen to stigma: This is pollination. For fertilization to occur, the pollen grains must first be transferred from the anther to the receptive surface of the stigma. Triple fusion: Once the pollen tube reaches the embryo sac, it releases two male gametes. One fuses with the egg (syngamy), and the other fuses with the two polar nuclei. This second process is triple fusion, which completes the act of double fertilization. Degradation of antipodals: This is an early post-fertilization event. After fertilization has taken place, the synergids and the antipodal cells, having fulfilled their roles, begin to degenerate. Fruit formation: This is a late post-fertilization event. As the seeds develop from the ovules, the ovary ripens and transforms into the fruit.

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113. (4)

Biological names are generally in Latin and written in italics. They are Latinised or derived from Latin irrespective of their origin. The first word denoting the genus starts with a capital letter. The specific epithet starts with a small letter. Biological names when handwritten are separately underlined.

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114. (3)

Detritivores (e.g., earthworm) break down detritus into smaller particles. This process is called fragmentation. This process physically occurs on the surface of the soil or within the topsoil where these organisms live and feed. The raw material for decomposition is detritus, which is defined as dead organic matter, including fallen leaves, dead animals, and other organic waste. The release of inorganic nutrients into the soil is known as mineralization, which is one of the final steps of decomposition. The process shown, fragmentation, primarily increases the surface area of the detritus for microbial action.

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115. (3)

Lactose is the substrate for the enzyme beta-galactosidase and it regulates switching on and off of the *lac* operon. Hence, it is termed as inducer. The repressor of the operon is synthesised (all-the-time – constitutively) from the *i* gene. In the presence of an inducer, such as lactose or allolactose, the repressor is inactivated by interaction with the inducer. Regulation of *lac* operon by repressor is referred to as negative regulation.

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116. (3)

In a dicot stem, each vascular bundle is conjoint, open, and with endarch protoxylem.

(NCERT Ed. 2025-26, 11th, Page No. 76)

117. (2)

Cuckoo laying eggs in Crow's nest	Brood parasitism	One benefitted, another harmed
Orchid growing as an epiphyte on a mango branch	Commensalism	One benefitted, another unaffected
Fig and pollinator wasp	Mutualism	Both benefitted
Visiting Flamingoes and resident fishes in some shallow South American lakes	Competition	Both lose

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118. (3)

The cell walls of fungi are composed of chitin and polysaccharides. With the exception of yeasts which are unicellular, fungi are filamentous. The network of hyphae is known as mycelium. Most fungi are heterotrophic and absorb soluble organic matter from dead substrates and hence are called saprophytes.

(NCERT Ed. 2025-26, 11th, Page No. 16)

119. (1)

Less than 2 per cent of the genome codes for proteins. Chromosome 1 has most genes (2968), and the Y has the fewest (231).

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120. (4)

The plant growth regulators (PGRs) are small, simple molecules of diverse chemical composition. They could be indole compounds (indole-3-acetic acid, IAA); adenine derivatives (N^6-furfurylamino purine, kinetin), derivatives of carotenoids (abscisic acid, ABA); terpenes (gibberellic acid, GA_3) or gases (ethylene, C_2H_4).

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121. (2)

In gymnosperms, the male and female gametophytes do not have an independent existence. Strobili or cones are distinct compact structures found in certain pteridophytes (like *Selaginella* and *Equisetum*) and gymnosperms. Mosses do not form cones; their sporophyte consists of a foot, seta, and capsule. The "precursor to seed habit" (retention of the female gametophyte on the parent sporophyte) is only seen in heterosporous pteridophytes (like *Selaginella* and *Salvinia*). Most pteridophytes are homosporous and do not show this trait.

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122. (3)

In most plants, by the time the fruit develops from the ovary, other floral parts degenerate and fall off. However, in a few species such as apple, strawberry, cashew, etc., the thalamus also contributes to fruit formation. Such fruits are called false fruits. Most fruits however develop only from the ovary and are called true fruits. Although in most of the species, fruits are the results of fertilisation, there are a few species in which fruits develop without fertilisation. Such fruits are called parthenocarpic fruits. Banana is one such example.

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123. (1)

In mustard, tap root system, alternate phyllotaxy, reticulate venation, actinomorphic flower, superior ovary, syncarpous condition, parietal placentation are found.

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124. (3)

The two chains of DNA helix have anti-parallel polarity. It means, if one chain has the polarity 5'→3', the other has 3'→5'. This means if one end of the double helix has a phosphate group (5' end) on one strand, it will have a hydroxyl group (3' end) on the other strand. The strands are complementary, meaning the sequence of one dictates the sequence of the other through specific base pairing.

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125. (2)

The vascular bundles are surrounded by a layer of thick walled bundle sheath cells.

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126. (3)

Gynaecomastia is a feature of Klinefelter's Syndrome.

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127. (1)

Glucose	Favored substrate for respiration
Fats	Have RQ below 1
Proteins	Will degrade into amino acids for respiration
Pyruvic acid	Can enter fermentation as well as Krebs' cycle depending upon cellular need

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128. (2)

Plum	Half-inferior ovary
Citrus	Polyadelphous stamens
Canna	Asymmetrical flower
Cassia	Imbricate aestivation

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129. (3)

The 'bakanae' (foolish seedling) disease of rice seedlings is caused by a fungal pathogen *Gibberella fujikuroi*.

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130. (1)

In an expanding population, the number of pre-reproductive individuals is more than that of reproductive and post-reproductive individuals. Sex ratio is a population attribute.

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131. (4)

Spirogyra is placed in algae (kingdom Plantae).

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132. (1)

The endomembrane system include endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles. Since the functions of the mitochondria, chloroplast and peroxisomes are not coordinated with the above components, these are not considered as part of the endomembrane system.

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133. (4)

The DNA can be digested by the enzyme DNase.

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134. (3)

In an isobilateral leaf, the stomata are present on both the surfaces of the epidermis. All tissues except epidermis and vascular bundles constitute the ground tissue. Pith is large and well developed in monocot root.

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135. (2)

One turn of Krebs' cycle produces 1 GTP (equivalent to 1 ATP), 3 NADH + H^+ (produce 3 × 3 = 9 ATP via oxidative phosphorylation) and 1 $FADH_2$ (produces 2 ATP via oxidative phosphorylation). Therefore, total ATP = 12.

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[Contd...

136. (1)

The variation in gene frequencies within a population occurring by chance rather than by natural selection is defined as genetic drift. It is a random process that can lead to significant changes in the gene pool, especially in small, isolated populations. Random mating maintains genetic equilibrium rather than causing chance variations. Genetic load refers to the presence of unfavorable genetic material in a population. Genetic flow involves the transfer of genetic material between populations through migration.

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137. (3)

Bone is a specialized connective tissue that serves as the primary structural frame for the body, supporting and protecting softer tissues and organs. Osteocytes are located in spaces called lacunae, not lamellae. Bone is characterized by a hard and non-pliable ground substance rich in calcium salts. Statement (4) is incorrect because it is cartilage, not bone, that is rich in chondroitin salts; bone strength comes from calcium salts and collagen fibers.

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138. (4)

Vinblastine	Drug
Anthocyanin	Pigment
Abrin	Toxin
Morphine	Alkaloid

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139. (1)

In the human respiratory system, the conducting part begins at the external nostrils and extends down to the terminal bronchioles. Expiration occurs when the intra-pulmonary pressure is higher than the atmospheric pressure, forcing air out. The cartilaginous rings supporting the trachea and bronchi are incomplete (C-shaped), not complete. Vital Capacity is the sum of IRV, ERV, and TV; it does not include the Residual Volume (RV).

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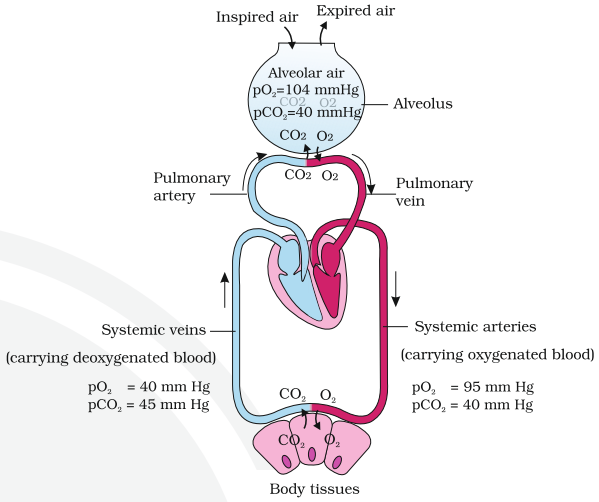
140. (2)

Malignant tumors are characterized by their ability to show metastasis, where cancerous cells slough off the original tumor and reach distant sites via the blood to start new tumors. Cancer cells lose the property of contact inhibition. Malignant tumors typically grow rapidly and cause extensive damage. Malignant tumors are inherently cancerous, unlike benign tumors.

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141. (2)

In the provided diagram of gas exchange, the blood vessel marked as B represents a systemic artery carrying oxygenated blood from the heart to the body tissues. In oxygenated blood, the partial pressure of oxygen (pO_2) is 95 mm Hg and the partial pressure of carbon dioxide (pCO_2) is 40 mm Hg.



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142. (3)

The lub sound, or the first heart sound, is produced by the closure of the tricuspid and bicuspid (mitral) valves. This occurs at the beginning of ventricular systole when the increasing pressure in the ventricles forces these valves shut to prevent backflow into the atria.

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143. (4)

GFR depends on blood pressure, blood flow, and hormonal control (renin/RAAS).

Nutrient levels (glucose, amino acids) do not influence filtration pressure, so they do not affect GFR.

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144. (2)

The correct pairing is *Pristis* — Saw fish. *Petromyzon* is the Lamprey, while the Hagfish is Myxine. *Carcharodon* is the Great White Shark, whereas the Dogfish is *Scoliodon*. *Exocoetus* is the Flying fish, while the Fighting fish is *Betta*.

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145. (1)

The development of the cockroach (*Periplaneta americana*) is termed paurometabolous, meaning it develops through a series of nymphal stages. This is accurately explained by the fact that the nymph looks very similar to the adult and grows by moulting approximately 13 times to finally reach the adult form.

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146. (1)

Competitive inhibitors resemble the substrate and bind to the active site of the enzyme. In the diagram, the substrate is shown as a triangular shape that fits into the V-shaped active site. Now compare the inhibitors:

P → contains a triangular portion, resembling the substrate shape → can compete for the active site. S → diamond shape with pointed ends similar to the substrate geometry → also likely to fit the active site.

Q, R, T → shapes do not resemble the substrate, so they are less likely to act as competitive inhibitors.

Therefore, the best pair of competitive inhibitors is: P and S.

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147. (2)

Erythroblastosis foetalis occurs specifically when the mother is Rh negative and the foetus is Rh positive.

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148. (4)

Physalia	Cnidoblast
Spongilla	Spicules
Apis	Chitinous exoskeleton
Asterias	Calcareous ossicles

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149. (4)

'Saheli' is unique because it is a 'once-a-week' pill with very few side effects. Statement A is incorrect because it is an oral pill, not an IUD. It is a non-steroidal preparation and does not contain estrogen. It was developed by the Central Drug Research Institute (CDRI) in Lucknow, not the ICMR.

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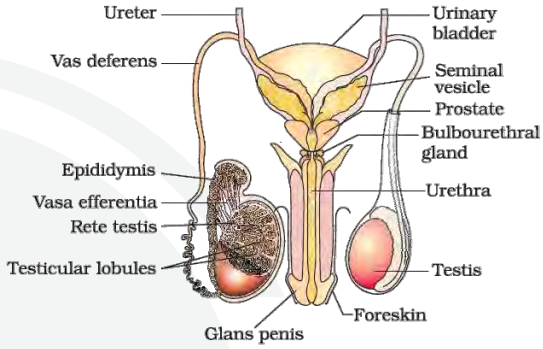
150. (4)

Adult human hemoglobin is a quaternary structure consisting of 4 subunits (two alpha and two beta chains). The first amino acid is the N-terminal, not C-terminal. Tertiary structures are stabilized by various bonds like hydrogen and disulfide bonds, not just peptide bonds. While Collagen is the most abundant protein in the animal world, RuBisCO is the most abundant in the entire biosphere. Secondary structures are mainly determined by hydrogen bonding.

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151. (2)

In the male reproductive system diagram, the part labelled B represents the seminal vesicles. These glands contribute the maximum volume (about 60-70%) to the seminal plasma and are the primary source of fructose, which provides energy for sperm motility.



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152. (2)

Multiload 375 is a copper-releasing IUD. Progestasert and LNG-20 are hormone-releasing IUDs. Lippes loop is a non-medicated IUD.

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153. (2)

Marijuana is a cannabinoid drug obtained from the leaves and inflorescences of the plant *Cannabis sativa.*

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154. (3)

Downstream processing refers to the stages that occur after the biosynthetic stage (the actual production in the bioreactor) is complete. It involves the separation and purification of the desired product, formulation with preservatives, and rigorous quality control testing.

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155. (3)

Aschelminthes are circular in cross-section, possess an excretory tube and pore, and have a complete alimentary canal with a muscular pharynx. While they are dioecious, females are typically longer than males. Fertilization is internal and development can be indirect.

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156. (3)

In the human body, macrophages act as an "HIV factory". After infection, the virus enters the macrophage where its RNA replicates into DNA, allowing the cell to continuously produce and release new HIV particles without being immediately destroyed.

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157. (2)

The presence of glucose and ketone bodies in urine is a classic clinical indicator of Diabetes Mellitus. The kidney can concentrate urine up to 4 times, not 10 times. Dialysis fluid is designed to have the same composition as plasma but lacks nitrogenous wastes to allow them to diffuse out of the blood. Sweat's primary function is cooling, though it removes small amounts of wastes.

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158. (3)

Trophoblast	Outer layer that gets attached to the endometrium
Inner cell mass	Differentiates into the embryo
Blastocyst	Arranged into trophoblast and inner cell mass
Morula	Contains 8 to 16 blastomeres

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159. (4)

α-1-antitrypsin is used to treat emphysema, not rheumatoid arthritis.

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160. (2)

Cloaca	A common chamber used to pass faeces, urine, and sperms
Bidder's Canal	Present in the kidney of male frogs to help in sperm passage
Mesorchium	A double fold of peritoneum that attaches testes to the kidneys
Conus Arteriosus	A sac-like structure on the ventral side of the heart where the ventricle opens

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161. (2)

The corpus luteum is formed and maintained by the action of Luteinizing Hormone (LH). In a non-pregnant cycle, it degenerates as LH levels fall. A continuous hypersecretion of LH would artificially maintain the corpus luteum, causing continued progesterone secretion and preventing menstruation.

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162. (3)

Lyases are enzymes that catalyze the removal of groups from substrates by mechanisms other than hydrolysis, often leaving double bonds.

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163. (4)

Adaptive Radiation	Darwin's Finches
Convergent Evolution	Wings of butterfly and bird
Divergent Evolution	Thorns of *Bougainvillea* and tendrils of *Cucurbita*
Anthropogenic Evolution	Selection of drug resistant bacteria

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164. (4)

In female cockroaches, the 7th sternum is boat-shaped and, along with the 8th and 9th sterna, forms the genital pouch. The head is formed by 6 segments, not 8, anal styles are present only in males and tegmina are opaque and leathery, not transparent.

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165. (2)

The enzyme that breaks down cellulose is cellulase; therefore, the enzyme that breaks down chitin is chitinase. Lysozyme acts on bacterial cell walls.

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166. (2)

Innate Immunity	Non-specific, present at birth
Acquired Immunity	Memory-based, pathogen-specific
IgA Antibodies	Transferred from mother to child during lactation
Passive Immunity	Transfer of preformed antibodies

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167. (3)

Consequence of inserting the gene at *Pvu*II rop gene gets disrupted. Both antibiotic resistance genes remain intact.

Therefore, transformed cells containing the recombinant plasmid will still be:

Ampicillin resistant

Tetracycline resistant

Because the selectable markers are not disrupted, recombinant and non-recombinant plasmids behave the same on antibiotic plates.

Thus, antibiotic selection cannot distinguish recombinants.

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[Contd...

168. (4)

While several treatments exist, the only permanent cure for ADA deficiency is the introduction of the functional ADA gene into cells at early embryonic stages. Periodic infusion and enzyme therapy require repeated treatments throughout life.

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169. (4)

The human skull is dicondylic, having two occipital condyles that articulate with the atlas.

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170. (3)

Thyocalcitonin (TCT) is a hypocalcemic hormone, meaning it decreases blood calcium levels. Insulin does lower glucose, glucocorticoids stimulate metabolic processes like gluconeogenesis, and catecholamines are emergency hormones.

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171. (3)

Cenozoic is widely known as the "Age of Mammals" due to their dominance during this era. Analogous organs are the result of convergent evolution, where different structures evolve to perform similar functions in similar environments.

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172. (3)

GIFT (Gamete Intra-Fallopian Transfer) involves the transfer of an ovum (collected from a donor) into the fallopian tube of another female who cannot produce one but can provide a suitable environment for fertilization.

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173. (3)

Regarding *Bt* toxin, the correct statement is that the activated toxin binds to the surface of midgut epithelial cells, creating pores that lead to cell swelling and death of the insect.

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174. (3)

The granular bodies found in the cell body and dendrites of a neuron are called Nissl's granules. They are composed of rough endoplasmic reticulum and free ribosomes, serving as sites for protein synthesis.

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175. (2)

The correctly matched pair is Hind II — First restriction endonuclease. Thermostable DNA polymerase (*Taq)* comes from *Thermus aquaticus*, not *Agrobacterium*. A Bioreactor is a vessel for processing, not an undifferentiated mass. *Ori* is the origin of replication, while recognition sites are specific DNA sequences for restriction enzymes.

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176. (2)

Secretin is the hormone that acts on the exocrine pancreas to stimulate the secretion of water and bicarbonate ions. Gastrin stimulates gastric acid, and CCK stimulates pancreatic enzymes and bile.

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177. (3)

Ball and Socket	Between Humerus and Pectoral Girdle
Hinge Joint	Knee Joint
Pivot Joint	Between Atlas and Axis
Saddle Joint	Between Carpal and Metacarpal of thumb

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178. (2)

Agrobacterium tumefaciens is known as a "natural genetic engineer" or biological vector because it has the natural ability to deliver a piece of its DNA (T-DNA) into plant cells. This makes it an ideal choice for delivering specific genes, such as herbicide resistance, into crops.

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179. (1)

Immediately after implantation, the inner cell mass differentiates into an outer layer called ectoderm and an inner layer called endoderm. The mesoderm appears soon after between these two layers.

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180. (3)

In Ctenoplana (a Ctenophore), digestion is both extracellular and intracellular. Ctenophores generally have external fertilization and indirect development, not direct one. Most ctenophores pass through a characteristic larval stage called the cydippid larva.

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